ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

TECHNOLOGY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 8 Technology paper 1 · practice a.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 B ✓ 1.4 D ✓ 1.5 A ✓ 1.6 C ✓ 1.7 C ✓ 1.8 A ✓ 1.9 D ✓ 1.10 B ✓ [10]

QUESTION 2

2.1 A beam bridge ✓ B arch bridge ✓ C suspension bridge ✓ D truss bridge ✓ (4)

2.2 A suspension bridge ✓ – it can span very long distances with few supports in the water ✓ (2)

2.3.1 Tie ✓ (1)

2.3.2 Strut ✓ (1)

2.3.3 The load pushes member Y towards the wall, so Y is being squashed (in compression) ✓ (1)

2.4 (a) Bending ✓ (b) Shear ✓ (c) Torsion ✓ (3)

2.5 The point where all the weight of an object seems to act ✓. Any TWO: make the base wider ✓; keep heavy parts low (lower the centre of gravity) ✓; anchor or bolt it down; use guy ropes or braces (3)

2.6 Concrete is strong in compression but weak in tension ✓. When a beam bends, the bottom part is stretched (in tension) ✓; the steel bars are strong in tension and stop the concrete from cracking ✓. (3)

[18]

QUESTION 3

3.1 Thermoplastics soften when heated and can be reshaped many times ✓, e.g. acrylic, PVC, polythene, PET ✓. Thermosetting plastics set hard permanently when first heated and cannot be reshaped ✓, e.g. Bakelite, melamine, epoxy resin ✓. (4)

3.2 Any ONE process ✓ with a product ✓: injection moulding – bottle caps, chairs; extrusion – pipes; blow moulding – bottles; vacuum forming – food trays; rotational moulding – water tanks (2)

3.3 They show the type of plastic ✓, so that plastics can be sorted correctly for recycling ✓. (2)

3.4 Ferrous metals contain iron (and usually rust), e.g. steel, cast iron ✓; non-ferrous metals contain no iron, e.g. copper, aluminium, brass ✓ (2)

3.5 Any TWO: work in a well-ventilated area because of toxic fumes ✓; wear safety goggles and gloves ✓; do not touch hot surfaces or hot wire; keep plastic away from open flames; switch off equipment after use (2)

[12]

QUESTION 4

4.1.1 Clockwise ✓ (the idler makes C turn in the same direction as A) (1)

4.1.2 $\text{VR} = \text{teeth on driven} \div \text{teeth on driver} = 45 \div 15 \checkmark = 3:1 \checkmark$ (2)

4.1.3 $N_C = 90 \times {}^{15}/_{45} \checkmark = 30 \text{ r/min} \checkmark$ (2)

4.1.4 It is an idler gear: it makes the driven gear turn in the same direction as the driver (it does not change the velocity ratio) ✓ (1)

4.2 (a) $N = 1\,200 \times {}^{40}/_{160} \checkmark = 300 \text{ r/min} \checkmark$ (b) $4:1 \checkmark$ (c) Cross the belt ✓ (4)

4.3 (a) Any ONE: guitar tuning pegs; large speed reduction in a winch or gate motor ✓ (b) Any ONE: hand drill; car differential; anything that changes the direction of rotation through 90° ✓ (c) Any ONE: car steering; sliding gate opener; pillar drill feed ✓ (3)

4.4 $\text{Effort} = 800 \div 4 = 200 \text{ N} \checkmark; \text{MA} = 800 \div 200 = 4 \checkmark$ (2)

4.5 A low gear gives a speed reduction ✓; the pedals turn more times for each turn of the wheel ✓, so less force is needed on the pedals to climb the hill (more turning force at the wheel) ✓. (3)

[18]

QUESTION 5

A	B	Lamp
0	0	0
0	1	0
1	0	0
1	1	1

5.1 Truth table correct as shown ✓✓✓; AND logic ✓ (4)

5.2 (a) OR logic ✓ (b) Any ONE: a doorbell with buttons at the front and back doors; a light that can be switched on from two places; a car's interior light that comes on when any door opens ✓✓ (3)

5.3 Any suitable example, e.g. a guillotine or press that only works when BOTH hands press two buttons ✓, so the operator's hands cannot be under the blade ✓ (2)

5.4 (a) To limit the current so the LED does not burn out ✓ (b) It does not light (an LED lets current through in one direction only) ✓ (c) Any ONE: uses less electricity; lasts much longer; does not get as hot ✓ (3)

5.5 (a) In the parallel circuit ✓ (b) Series: the other bulb goes out ✓ because the circuit is broken ✓; parallel: the other bulb stays on ✓ (4)

5.6 A fuse ✓ (accept a circuit breaker) (1)

[17]

TOTAL: 75

END OF MARKING GUIDELINE